

EEE 120
Lab 1 Answer Sheet

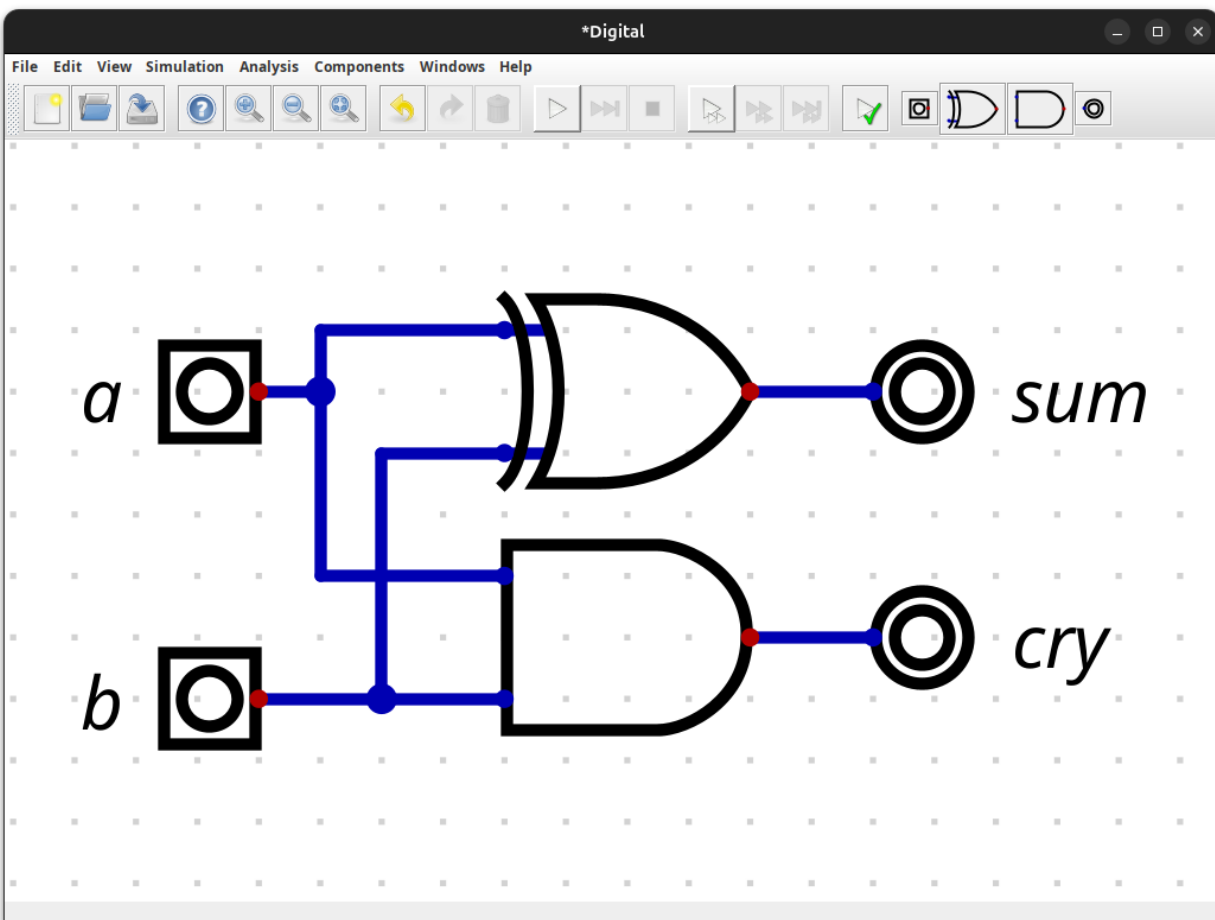
Half Adder, Full Adder, 4-bit Incrementer and Adder

Name: Dylan Shah

Instructor/Time: Dr. Steve Millman Tu Th 3:00 PM - 4:15 PM Date: January 13, 2026

Task 1-1: Build and Test the 1-Bit Half-Adder

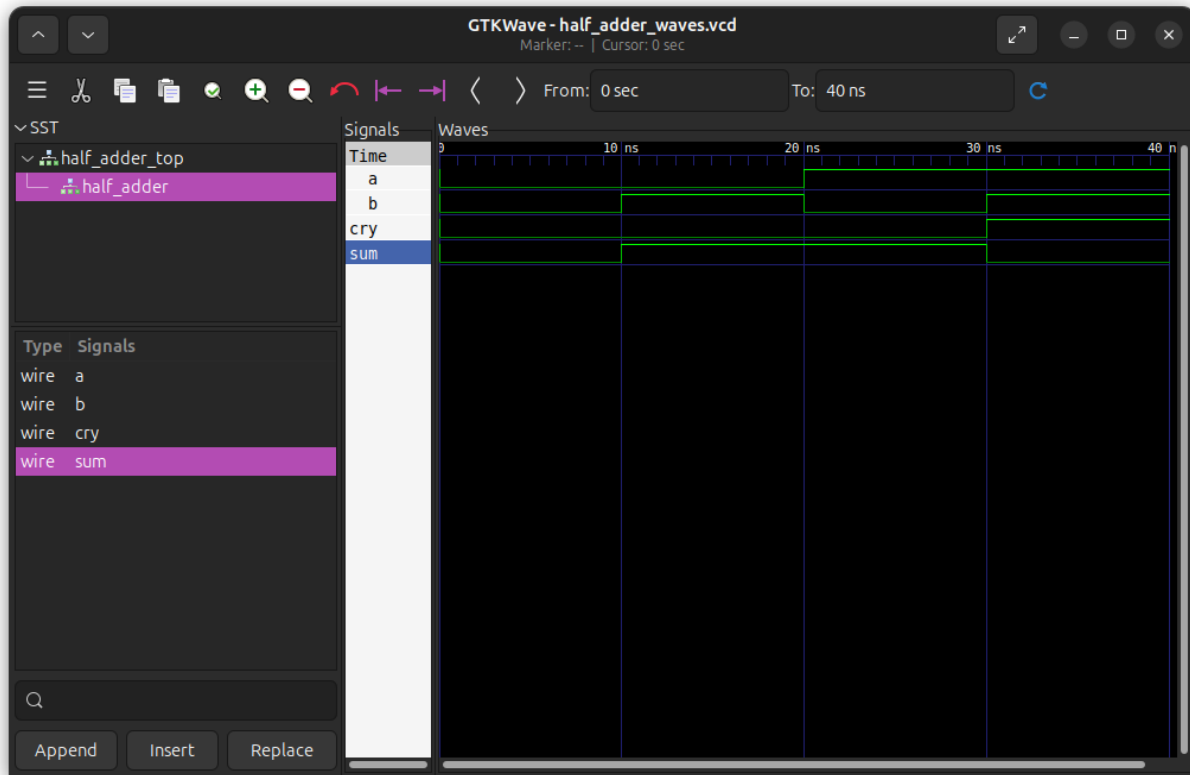
Include a picture of your circuit in Digital here:



Please comment on the single biggest issue you were facing when designing the circuit.

I did not face any issues while creating this circuit.

Include a picture of your waveform (timing diagram) here:



Did the circuit behave as expected? If no, what was wrong?

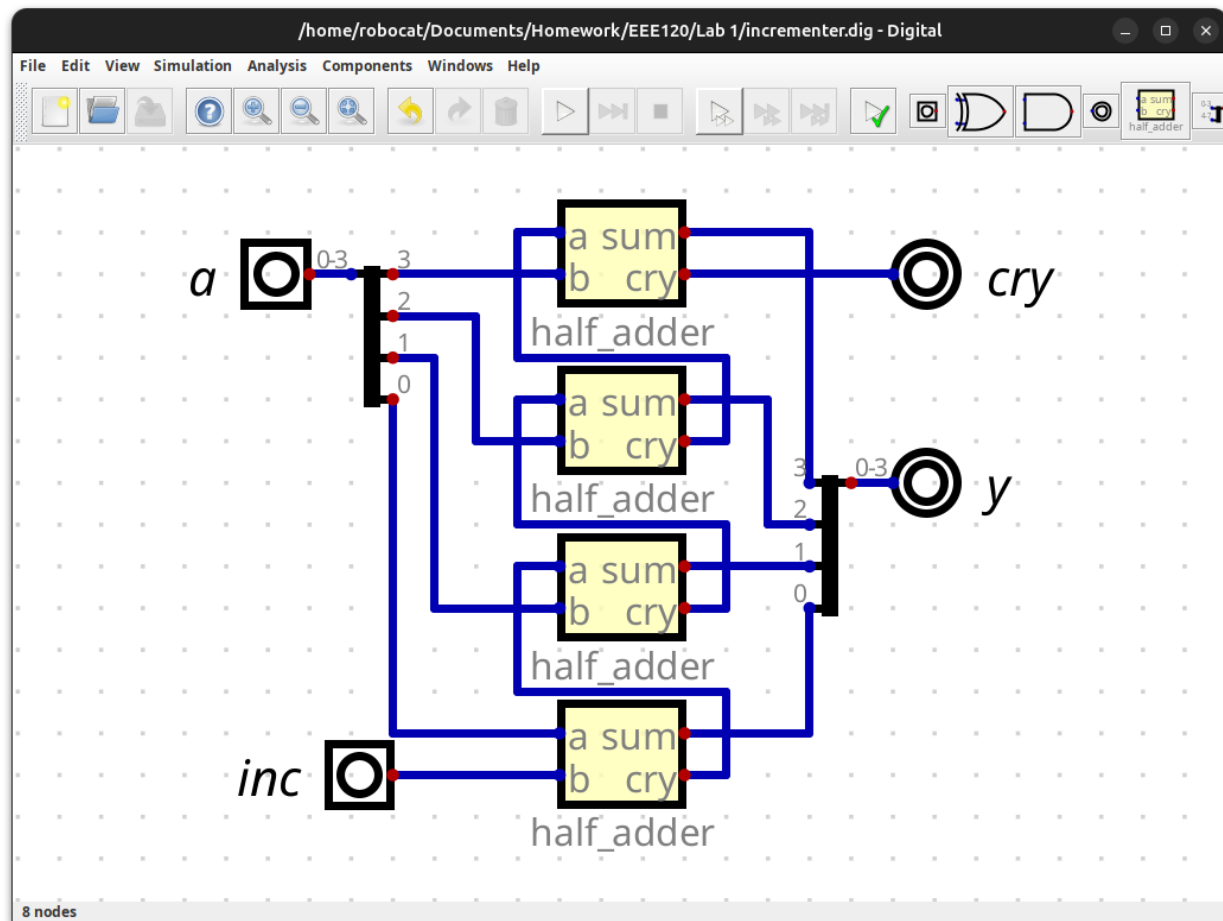
Yes, the circuit behaved as expected.

Please comment on the single biggest issue you were facing when simulating the circuit.

I did not face any issues simulating the circuit.

Task 1-2: Build and Test a 4-Bit Increment Circuit

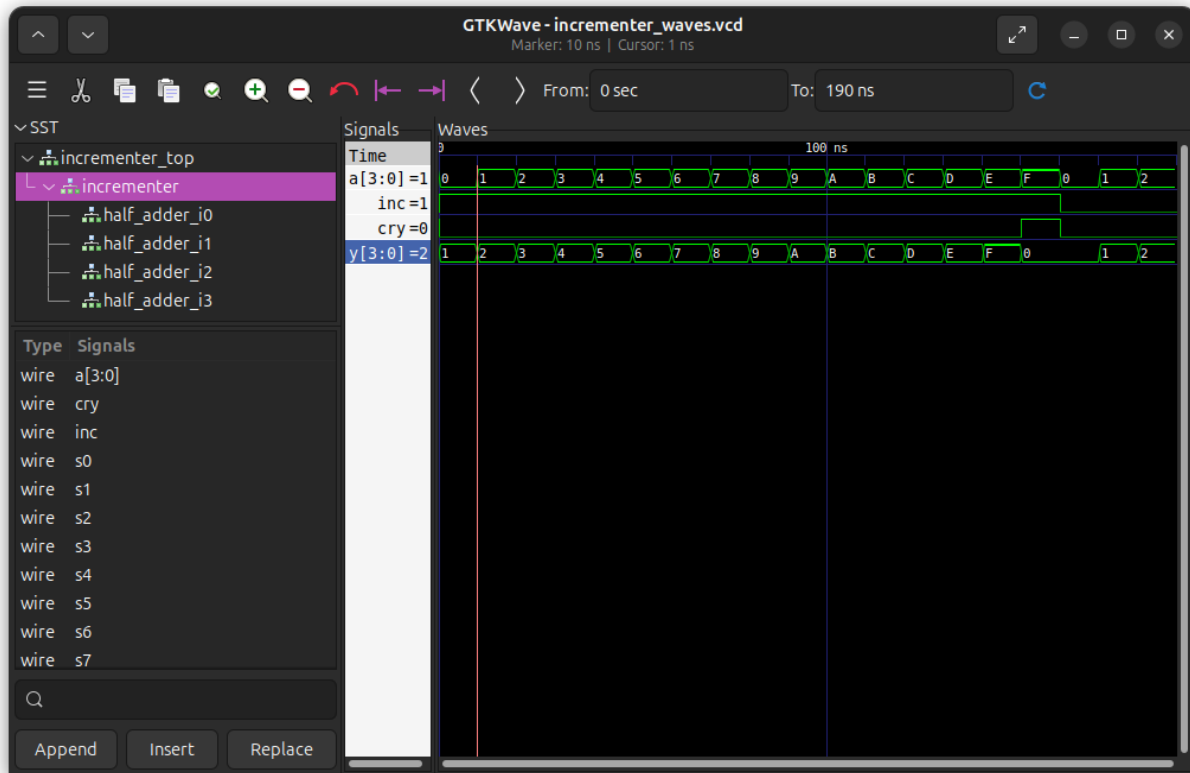
Include a picture of your circuit in Digital here:



Please comment on the single biggest issue you were facing when designing the circuit.

I kept connecting wires I wanted to connect to the sum to cry on the half adder, and vice versa.

Include a picture of your waveform (timing diagram) here:



Did the circuit behave as expected? If no, what was wrong?

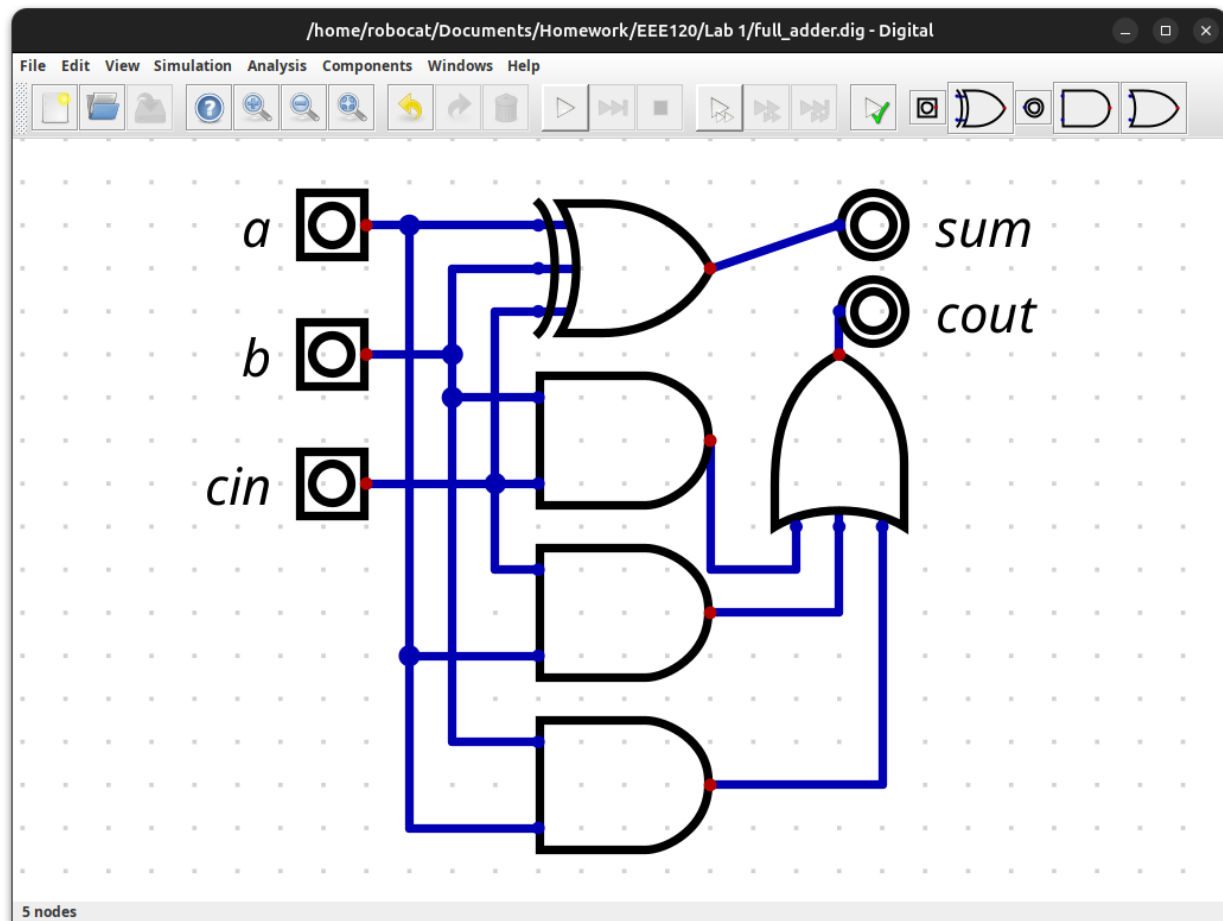
The circuit is working as expected.

Please comment on the single biggest issue you were facing when simulating the circuit.

I did not face any issues simulating the circuit.

Task 1-3: Build and Test a 1-bit Full Adder

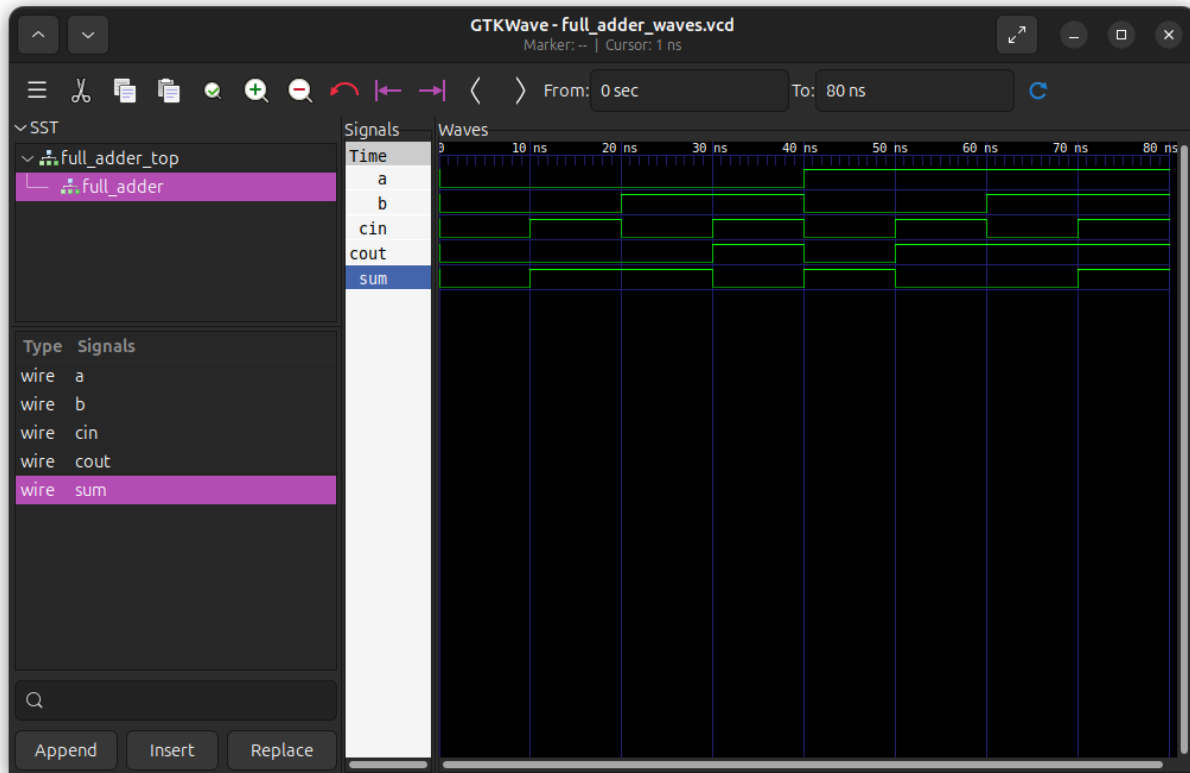
Include a picture of your circuit in Digital here:



Please comment on the single biggest issue you were facing when designing the circuit.

For some reason, I wanted to make the circuit as small as possible, which resulted in me routing the wires to the and gates in ways that I did not want to do, as there were already wires in the places that I would have preferred to put them if I gave myself a reasonable amount of space. I have no clue why I did this to myself.

Include a picture of your waveform (timing diagram) here:



Did the circuit behave as expected? If no, what was wrong?

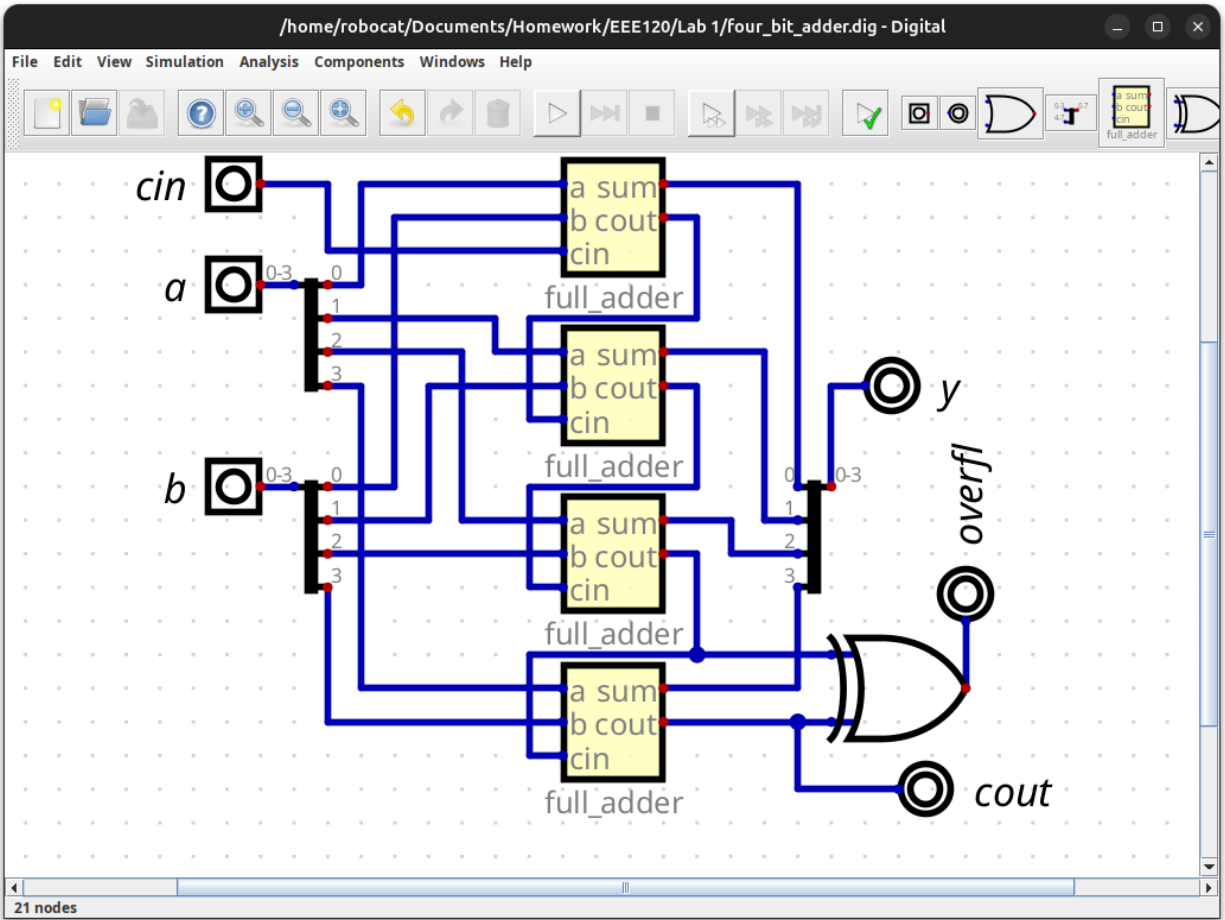
Yes, the circuit behaved as expected.

Please comment on the single biggest issue you were facing when simulating the circuit.

I forgot to export the verilog file, and then got errors when I was simulating (the non-existent file). Once I actually exported the file, everything worked.

Task 1-4: Build and Test a 4-Bit Full Adder

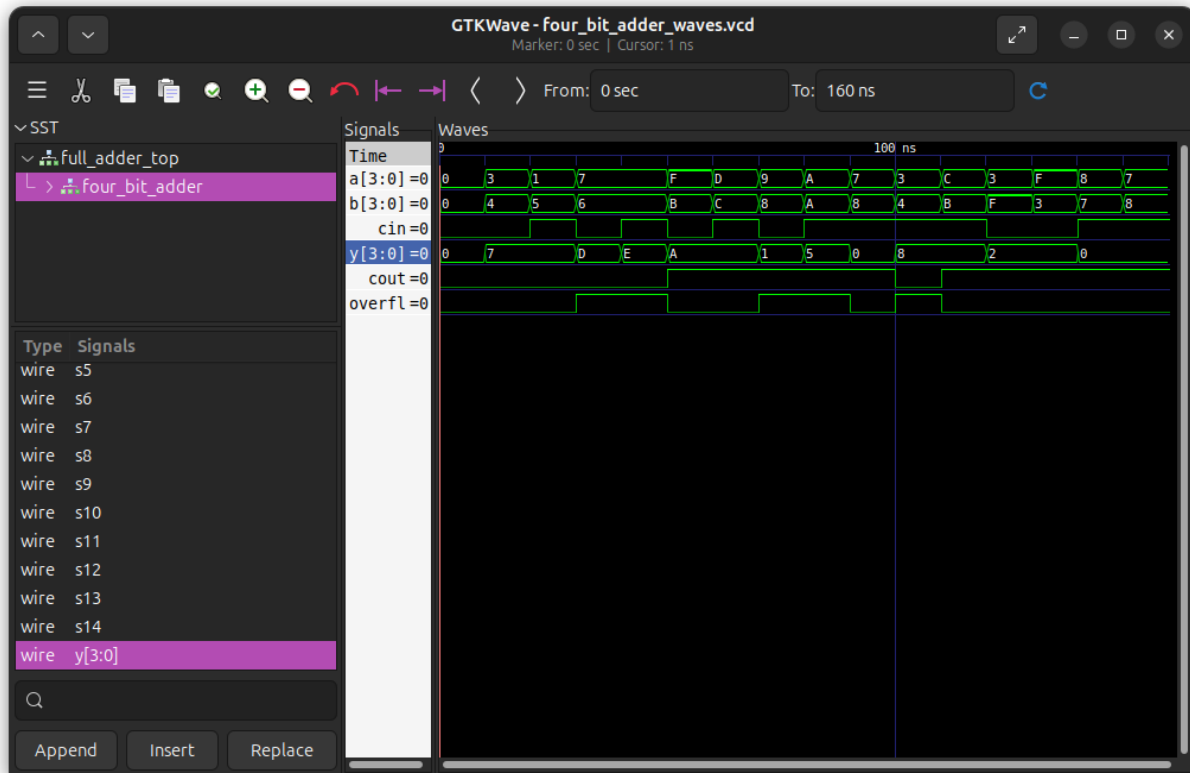
Include a picture of your circuit in Digital here:



Please comment on the single biggest issue you were facing when designing the circuit.

I did not face any issues designing this circuit.

Include a picture of your waveform (timing diagram) here:



Which tests did you perform and why? Use the following table to describe your test sequence. You need to make sure to perform enough tests to check the circuit for possible incorrect behavior. Each row of the first column corresponds to a row of stimulus from your four_bit_adder_stim.txt file. Note that you only need to use as many tests as needed. Extra space is provided for enthusiastic students. While it is expected that all of your tests will pass, if you run out of time, some may fail.

Test stimulus	Test motivation	Pass/Fail
0_0_0_0_0	Check for stuck-at-1 faults (this makes sure none of the wires were accidentally connected to power)	Pass
0_7_0_3_4	Verify $a + b$ is correct for unsigned values of a and b with no overflow and signed values of $+a$ and $+b$ where there is no overflow	Pass

Test stimulus	Test motivation	Pass/Fail
0_7_1_1_5	Verify $a + b + 1$ is correct for unsigned values of a and b with no overflow and signed values of $+a$ and $+b$ where there is no overflow	Pass
2_D_0_7_6	Verify $a + b$ is correct for unsigned values of a and b with no overflow and signed values of $+a$ and $+b$ where there is overflow	Pass
2_E_1_7_6	Verify $a + b + 1$ is correct for unsigned values of a and b with no overflow and signed values of $+a$ and $+b$ where there is overflow	Pass
1_A_0_F_B	Verify $a + b$ is correct for unsigned values of a and b with overflow and signed values of $-a$ and $-b$ where there is no overflow	Pass
1_A_1_D_C	Verify $a + b + 1$ is correct for unsigned values of a and b with overflow and signed values of $-a$ and $-b$ where there is no overflow	Pass
3_1_0_9_8	Verify $a + b$ is correct for unsigned values of a and b with overflow and signed values of $-a$ and $-b$ where there is overflow	Pass
3_5_1_A_A	Verify $a + b + 1$ is correct for unsigned values of a and b with overflow and signed values of $-a$ and $-b$ where there is overflow	Pass
1_0_1_7_8	Verify that $a + b + 1$ overflows when $a + b$ would not overflow where a and b are unsigned	Pass
2_8_1_3_4	Verify that $a + b + 1$ overflows when $a + b$ would not overflow where a and b are signed	Pass
1_8_1_C_B	Verify that $a + b + 1$ does not overflow when $a + b$ would overflow where a and b are signed	Pass

Test stimulus	Test motivation	Pass/Fail
1_2_0_3_F	Verify $a + b$ is correct for $+a$ and $-b$	Pass
1_2_0_F_3	Verify $a + b$ is correct for $-a$ and $+b$	Pass
1_0_1_8_7	Verify $a + b + 1$ is correct for $+a$ and $-b$	Pass
1_0_1_7_8	Verify $a + b + 1$ is correct for $-a$ and $+b$	Pass

Please comment on the single biggest issue you were facing when simulating the circuit.

I struggled to think of the last few test cases for the numbers

Task 1-5: Create a video and submit your report.

Record a short video showing your schematics in Digital and your waveforms in GTKWave. Be sure to show yourself in the video and show your screen. Explain how your circuit works – you need to convince the grader you did the lab and understand it! **Copy and paste the link to your video below. Make sure the link is working and pointing to the correct video. Remember to include the password if required. Do NOT upload your video to Canvas. It is recommended that you use Zoom to record to the cloud, pasting the link and password below.** If your circuit is not working as expected, explain in the video how it is not working and why you think it is not working.

Video Link:

<https://drive.google.com/file/d/142Ru2lwVI7un7MH3chllLdEeXE2rNmVy/view?usp=sharing>

At the beginning of your recording, say your name and the lab name. Be brief in your recording. Submit the completed template to Canvas.

Make sure all your files are in the Lab1 directory. Create a zip file of the Lab1 directory. Remember to turn in the zip file and your completed template on Canvas!

Do not include the video in the zip file! This makes the file very large and you run the risk of the zip file not uploading or taking so

long to upload that your submission will be late. Remember that the submission is dated at the time the upload completes, not when it starts!

LAB 1: LAB REPORT GRADE SHEET

Name _____

**NOTE: You submit the zip file in order to show your work.
If the zip file is not submitted you will receive a 0 for this lab!**

Instructor Assessment

Grading Criteria	Max Points	Points Lost
Description of Assigned Tasks, Work Performed &		
Task 1-1: Build and Test a 1-Bit Half-Adder	10	
Task 1-2: Build and Test a 4-Bit Increment Circuit	10	
Task 1-3: Build and Test a 1-Bit Full Adder	10	
Task 1-4: Build and Test a 4-Bit Full Adder	10	
Task 1-5: Create a video and submit your report.	10	
Lab Score (50 points total)	Points	
	Late Lab	
	Lab Score	